BRAIN 05

# Methylene blue

Blue thiazine dye; given into a vein as a drug (ProvayBlue) for acquired methemoglobinemia



**CLASS**
Small molecule (thiazine dye)

**STUDIED FOR**
Methemoglobinemia; memory, mood

**HUMAN EVIDENCE**
FDA approved; small brain studies

**US STATUS**
FDA approved (ProvayBlue) · brain use off-label

## STUDIED FOR

- **Methemoglobinemia** — HUMAN
- **Memory and attention** — HUMAN
- **Bipolar symptoms** — HUMAN
- **Mitochondria** — ANIMAL

**No energy trials found.** No published trial of methylene blue for energy or fatigue. A related compound, LMTM, showed no benefit in a Phase 3 Alzheimer's trial.

## HOW IT WORKS IN THE BODY

Methylene blue ▶ Leucomethylene ▶ Methemoglobin ▶ Hemoglobin

- **Label:** converted to leucomethylene blue, which turns methemoglobin back to hemoglobin
- **Label:** appears to be a potent reversible MAO inhibitor
- **Lab and animal:** reroutes electrons in mitochondria; protects brain cells in models
- **For energy in people,** the mechanism is not established

## EVIDENCE & WHAT'S KNOWN

### EVIDENCE

**FDA APPROVED**
LEVEL 5 OF 5

- FDA approved IV drug (2016) for methemoglobinemia
- Small brain studies: 26 and 37 people
- No trial for energy or fatigue

### NOT YET ESTABLISHED

- Energy effect in people
- Oral use for thinking or mood
- Repeat-dose brain effects

## QUICK PRACTICAL INFO

**HALF-LIFE**
About 24 hours (label)

**ROUTES STUDIED**
Into a vein (label); by mouth in brain studies

**PHARMACOKINETICS**
Mainly UGT conjugation; CYP1A2, 2C19, 2D6 in vitro; liver

**SOLD AS**
Prescription IV solution (ProvayBlue, generics)

**MONITORED IN TRIALS**
Methemoglobin; serotonin syndrome signs; hemolysis

**SPORT (WADA)**
Not on the 2026 list

**REGULATORY**

| | |
|---|---|
| FDA | Approved (ProvayBlue, 2016), IV only |
| FDA | Boxed warning: serotonin syndrome |
| EMA | Authorised: methaemoglobinaemia; colonoscopy dye |
| WADA | Not listed |


**REFERENCES**

1. FDA (DailyMed), ProvayBlue prescribing information, Jun 2025
2. Radiology, single-dose methylene blue and brain MRI, Nov 2016
3. Lancet, LMTM Phase 3 trial in Alzheimer's disease, Dec 2016



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